

the second line. It is then followed by display of 'Number of Cars' in the first line and '001' in the second line. If the parking lot is full, the message "Parking is Full, Sorry for Inconvenience" is displayed on LCD1.

When a car leaves the parking area and crosses the IR beam between IR LED2 and D2 at the exit gate, the vehicle count decreases by one. The LCD shows the number of cars in the parking lot along with "Thanks for Visiting" message.

Software

The program (parking.c) for the microcontroller is written in C and compiled using Keil software to generate the hex code. The program coding starts with '#include<reg51.h>' and '#include<string.h>' header files. The microcontroller port pins are defined using 'sbit' function for interfacing with the surrounding peripherals. The entry gate motor is controlled using 'sbit START_POINT=P1^3;'. code.

The LCD is initialised using the following code:

```
void lcdinit(void)
{
    lcdcmd(0x38);
    DelayMs(250);
    lcdcmd(0x0E);
    DelayMs(250);
    lcdcmd(0x01);
    DelayMs(250);
    lcdcmd(0x06);
}
```

```
DelayMs(250);
lcdcmd(0x08);
DelayMs(250);
}
```

Construction and testing

An actual-size, single-side PCB layout for the RFID-based automatic vehicle parking system is shown in Fig. 7 and its component layout in Fig. 8. Burn the hex code into the AT89S52 microcontroller using a suitable programmer and then mount the microcontroller on the PCB. Install IR LED1-D1 pair at the entry gate such that these face each other. Similarly, install IR LED2-D2 pair at the exit gate.

For testing, switch on the circuit, interrupt the infrared beam between IR LED1 and IR D1 with your hand or some other opaque object and then remove it, and place the tag near the reader. The LCD should show the message as described earlier in 'How this vehicle parking system works' section. An amount of Rs 100 should be deducted for every interruption of the IR beam. The card can be recharged by pressing the pushbutton switches (S2 and S3) provided in the circuit. Pressing switch S2 recharges the card with Rs 900 and pressing switch S3 recharges it with Rs 500.

Similarly, interrupt the IR beam at the exit gate. LCD1 should show the number of cars in the parking lot along with 'Thanks for Visit' message.

PARTS LIST

Semiconductors:	
IC1	- 7805, 5V regulator
IC2	- 7812, 12V regulator
IC3	- AT89S52 microcontroller
IC4	- LM358 dual-operational amplifier
RFID reader	- 9-pin 125kHz RFID reader
LED1, LED2	- 5mm light-emitting diode
IR LED1, IR LED2	- 5mm infrared transmitter diode
IRD1, IRD2	- 5mm infrared receiver diode
D1-D4	- 1N4007 rectifier diode
T1	- BC547 transistor
Resistors (all ¼-watt, ±5% carbon):	
R1	- 470-ohm
R4, R6, R8, R10	- 220-ohm
R2, R9	- 10-kilo-ohm
R3	- 100-ohm
R5, R7	- 1-mega-ohm
VR1, VR2	- 100-kilo-ohm preset
VR3	- 10-kilo-ohm preset
Capacitors:	
C1	- 1000µF, 35V electrolytic
C2	- 10µF, 16V electrolytic
C3, C4	- 22pF ceramic
C5	- 0.1µF ceramic
Miscellaneous:	
LCD1	- 16×2 LCD display module
CON1	- DC connector
CON2, CON3	- Two-pin berg strip male connector
CON4	- 9-pin berg strip female connector
PZ1	- Piezobuzzer
X ₁ , X ₂	- 11.0592MHz crystal
S1-S3	- Tactile switch
M1, M2	- DC motor

No amount should be deducted at the time of exit.

EFY note. The source code of this article is available on www.efymag.com website.

The author is founder and CEO of HBeonLabs, Greater Noida

PARKING.C

```
//Automatic RFID Based Vehicle Parking
System
#include<reg51.h>
#include<string.h>

sbit START_POINT=P1^3;
sbit TERMINATE_POINT=P1^4;

sbit S1=P3^2;
sbit S2=P3^3;

sbit rc1=P1^1;
sbit rc2=P1^2;

sbit BUZZPORT=P1^7;

sbit RS=P3^7;
sbit EN=P3^6;

void lcdinit()
{
    void lcdData(unsigned char l);
    void lcdcmd(unsigned char k);
    void buzzer(unsigned int time);
    void DelayMs(unsigned int count);
}
```

```
void display(unsigned char s,t,u);
void Welcome(unsigned char c[]);
void ConvertAndDisplay(unsigned int value,unsigned char c[]);
void dcMotor();

void main()
{
    unsigned char i=0,j=0,result=0;
    unsigned int count;
    unsigned char c[15];
    unsigned char d[]="42006B1BB8";

    signed int amount2=900;

    TMOD=0x20; // To configure the serial port at 9600 baud rate
    TR1=0xF0;
    SCON=0x50;
    TR1=1;

    S1=1;
    S2=1;
}
```

```
BUZZPORT=0;
START_POINT=0;
TERMINATE_POINT=0;

lcdinit();
Welcome("AUTOMATIC RFID","CAR PARKING");
DelayMs(1000);

while(1)
{
    known:
    while(S1==1 && S2==1);

    if(S2==0)
    {
        while(S2==0);
        if(count>0)
        {
            count--;
        }

        ConvertAndDisplay(count,"Thanks for Visiting");
    }
}
```